Pakistan Journal of Pharmacology
Vol.23, No.1, January 2006, pp.47-50

ROLE OF INDAPAMIDE ON HYPERTENSIVE LEFT VENTRICULAR HYPERTROPHY

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ABSTRACT:

Left ventricular hypertrophy (LVH) is a sequelae of sustained prolonged systemic hypertension Initially it is a compensatory mechanism but later on the left ventricular hypertrophy carries the risk of arrhythmias, congestive cardiac failure, angina, systolic as well as diastolic dysfunction, greater intensity and risk of myocardial infarction and cardiac rupture. The Framingham Heart Study and other studies have shown LVH as an independent risk factor for cardiovascular morbidity and mortality. Some classes of drug cause regression of LVH e.g. ACE inihibitor, AT_1 receptor blockers, calcium channel blockers, B blockers and methyldopa. The effect of diuretics on regression are equivocal. we gave indapamide 2.5mg to 7 pts with echocardiographic evidence of LVH and found that LVID (mm) (mean + SEM) decreased from 50.7 ± 1.2 to 50.20 ± 1.70, IVST (mean + SEM) decreased from 14.10 ± 0.4 to 11.9 ± 0.30, PWT (mean + SEM) decreased from 13.0 ± 0.3 to 11.2 ± 0.2 and LVMI (mean + SEM) decreased from 176.42 ± 6.13 to $145.25 + 6.04 g/m^2$

Keywords: Left vectricular hypertrophy indapamide, regression, LVMI

INTRODUCTION

The hypertensive subjects are prone to develop Left ventricular hypertrophy, which is associated with CCF (Devereux *et al.*, 1982), coronary heart disease (Kannel *et al.*, 1969), extrasystole, ventricular couplets and ventricular tachycardia (Mclenachan *et al.*, 1987). The Framingham Heart Study has demonstrated the association between LVH and Angina, myocardial infarction and sudden death (Messerli *et al.*, 1984). The left ventricular hypertrophy is a more powerful predictor than age, sex and ejection fraction, and as important as the number of stenosed coronary arteries. After reversal of LVH ventricular contraction is improved and systolic and diastolic functions are enhanced (Lavie *et al.*, 1992). Antihypertensive drugs that modulate the renin-angiotensin aldosterone system, sympathetic drive or intracellular free calcium concentration include angiotensin converting enzyme inhibitors, beta-blockers, centrally acting adrenergic drugs and calcium antagonists, which lead to regression (Katz, 1998). Diuretics remain first line agents in the treatment of hypertension. They not only effectively reduce blood pressure but also the cardiovascular morbidity and mortality. Indapamide, a once daily orally active sulfonamide diuretic is an effective anti-hypertensive agent and has shown evidence of

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Table-1
Effect of Indapamide on Blood Pressure

	On Day 0	After 6 Months
Systolic BP (mm Hg)	174.1 $\pm$ 2.9	146.0 $\pm$ 1.7
Diastolic BP (mm of Hg)	101.5 $\pm$ 1.2	85. ± 1.5

Mean ± standard error of mean

The echocardiographic parameters and LVMI obtained on day 0 and after 6 months of therapy are:

	On Day 0	After 6 Months
LVID (mm) (mean + SEM)	50.70 ± 1.20	50.20 ± 1.70
IVST (mean + SEM)	14.10 ± .40	11.9 ± 0.30
PWT (mean + SEM)	13.00 ± 0.30	11.2 ± 0.20
LVMI (mean + SEM)	176.42 ± 6.13	145.25 ± 6.04

LVID= left ventricular internal diameter in diastole.
IVST= Interventricular septal thickness
PWT= Posterior wall thickness
LVMI=left ventricular mass index (g/m^2)

regression of LVH in some studies. In order to see the effect of indapamide on regression of LVH we took the following study.

SUBJECTS AND METHODS

Seven patients were selected from cardiac OPD P.M.C Nawabshah between 7th January 2001 to 12th December 2001 and written consent were obtained from them. The inclusion criteria was patients having systolic blood pressure of 180mm of Hg or above or diastolic blood pressure between 95mm of Hg to 115 mm of Hg. Having echocardiographic evidence of LVH i.e. LVMI $\geq$130g/m^2 in male and $\geq$ 110g/m^2 in females aged between 29-75 years. Exclusion criteria was, secondary hypertension, coronary artery disease, heart failure, cardiac decompensation, patients having concomitant disease such as renal failure, Diabetes Mellitus or cardiomyopathy. Patients taking drugs like corticosteroid, digoxin, neuroleptics, antidepressant were also not included. M-mode echocardiogram was performed by using standard 3.5MHz transducer with Toshiba Sonolayer-LS SAL-38 as echocardiograph. The patients were kept in left lateral decubitus position which is standard echocardiographic technique. The transducer was placed parasternally in the 3rd and 4th intercostal space. The airless contact between transducer and the skin was ensured by using an ultrasonic gel as a coupling medium. The following parameters were obtained from the echocardiogram:

1. Interventricular septal thickness (mm) IVST
2. Posterior wall thickness (mm) PWT
3. Left ventricular internal diameter (mm) LVID
4. Left ventricular mass index (LVMI)

RESULTS

The age of seven patients ranged from 29-75 years (mean 54.6 years) and, all the patients had mild to moderate primary essential hypertension with mean diastolic blood pressure of 101.5 $\pm$ 1.2 mm of Hg and SBP 174.1 $\pm$ 2.9

mm of Hg Mean duration of hypertensoin was 9.1 $\pm$ 2.3 years. Indapamide induced significant reduction of both diastolic and systolic blood pressure

The mean reduction in supine blood pressure was 18.5 mm of Hg for diastolic and 28.1mm of Hg for systolic blood pressure. All the patients achieved normal blood pressure defined as a diastolic blood pressure below 90mm of Hg. The maximum effect on hypertension was observed within two weeks of treatment.

The posterior wall thickness decreased from 13 ± 0.3 (mm) to 11.2 ± .2mm (P<0.001) The interventricular septal thickness decreased from 14.10 ± .4 to 11.9 ± .0.3mm (P<.001) and left ventricular mass index decreased from 176.42 ± 6.13 to 145.25 ± 6.04 g/m^2 (P<.0001), a mean reduction of 16%. The left ventricular end diastolic diameter did not regressed significantly (from 50.7 ± 1.2 to 50.20 ± 170). However there was a significant relation between interventricular septal thickness and left ventricular mass index.

DISCUSSION

In our study the posterior wall thickness (PWT) decreased from 13.00 ± .30 to 11.2 ± .0.3mm (P<.001). Interventricular septal thickness decreased from 14.10 ± .40 to 11.9 ± .0.3mm (P<.001). The left ventricular mass index decrease was statistically significant. The LVMI decreased by 16%. The left ventricular end diastolic diameter decreased from 50.7 ± 1.2 to 50.20 ± 170. The reduction was not significant. There was a significant relation between interventricular septal thickness and left ventricular mass index. Sami and Haichin 1991 conducted study on eleven patients having left ventricular hypertrophy with indapamide. They found LVM (Mean ± SD) decreased from 254 ± 47(g) to 219 ± 46(g). The reduction was statistically significant (P<.003). The posterior wall thickness decreased from 11.1 (mm) to 10.1(mm). Left ventricular Internal diameter in diastole decreased from 51 ± 3(mm) to 49 ± 5(mm), interventricular septal thickness decreased from 12 ± 3 to 11 ± 1(mm) they did not reach statistical significance. LVMI decreased from 146 ± 22g/m^2 to 124 ± 22g/m^2 (P<0.003). The mechanism by which indapamide reduced left ventricular hypertrophy in our patients is not known. Devereux *et al* suggested reduction in peripheral resistance and after load may be the cause of regression with indapamide (Komajda and Carey *et al*., 1996). But other drugs such as aspresoline, minoxidil and prazosin which are also known to reduce peripheral resistance and cardiac after load have not been associated with similar decrease in left ventricular mass in patients with hypertension. Even some studies have shown that they might increase left ventricular hypertrophy despite reduction in blood pressure. These findings have prompted other theories for the reduction of left ventricular mass such as lowering of intracellular calcium ions (Homcy, 1998). Indapamide with regard to glucose and lipid profile is neutral in comparison with other diuretics. An ideal antihypertensive drug must have beneficial effects on other concomitant factors of cardiovascular morbidity and mortality (Leonetti, 2002). Long-term treatment with indapamide does not alter insulin or glucose metabolism reducing the risk of hyperglycemia reported with long term diuretic therapy (Tan *et al*., 1996). In addition the metabolic effects of indapamide are also characterized by the absence of deterioration of lipid metabolism, an important advantage when compared with other diuretic or B Blockers with no ISA (Intrinsic sympathetic activity). Indapamide does not deteriorate and may actually improve many cardiovascular risk factors, an interesting potential for long-term use in hypertensive patients. Indapamide with its ease of use, its widespread acceptance over many years and cardioprotective potential described here, needs to be investigated further to see if it can fulfill these expectation as an ideal inexpensive treatment (Campbell and Brackman, 1990).

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